30401 Mathematics and Statistics - Module 2 (Statistics) - BEMACS
FIRST PARTIAL - 60 minutes

Surname		Name		Student Number	

I declare I have read the Exam Rules and Honour Code and I commit to respect them.

Signature:

Please in your answers provide the fundamental steps of your proceedings.

Please label your answers and write clearly and with intelligible handwriting.

BEST OF LUCK!

Exercise 1 (13 points)

A company wants to evaluate the effectiveness of the promotional campaigns for its products at stores that offer large discounts on the company's products. The company conducts a survey of a sample of stores where equivalent promotions were applied. The survey results are contained in the **Campaign** dataframe in the **Campaign.Rdata** file which includes the following variables:

Id: Store identifier

Location: Location of the store (Hinterland, Peripheral, Semi-Central, Central)

Loyalty: Variable measured in classes indicating the store's level of customer loyalty

Revenues: Numerical indicator of the standard (that is in the absence of promotional campaigns) profitability of the store in relative terms

Rewards: Numerical indicator of the success - in relative terms - of the promotional campaign on the company's products in the specific store.

Costs: Numerical indicator of the costs - in relative terms - of the promotional campaign.

Sales: Numerical indicator of the effectiveness of the campaign based on the purchases made by customers at the store (in relative terms).

a) (1pt*) What is the store **Location** most represented in the sample?
   - ◻ Hinterland ◻ Peripheral ◻ Semi-Central ◻ Central

b) (1pt*) What are the percentages of stores located in the city centre (**Location** = Central) and hinterland (**Location** = Hinterland) respectively?

c) (1pt*) Consider the variables **Loyalty** and **Location**. Report the proportion of stores with the highest loyalty level (i.e. **Loyal** = [80,100]) among all the stores with central location and among all the stores located in the Hinterland respectively. Compare the two proportions obtained.

d) (1pt*) What is the sample mean **Revenues** of the stores in the sample?
   - ◻ 1057.6 ◻ 984 ◻ 410.2 ◻ 1682.9

e) (2pt**) Consider the boxplot representing the sample distribution **Revenues**.

   i. Indicate what the extremes of the box and the end points of the lower whisker of the boxplot represent.

   ii. Report their numerical values, clarifying what are the quantities underlying your answer.

f) (2pt*) Propose a graphical representation to compare the conditional distributions of the standard profitability of the stores (variable **Revenues**) according to their location with respect to the city centre (**Location**).

   i. Report the graph obtained (a simple sketch, just to understand what your comments refer to).

   ii. Briefly comment on the symmetry of each of the distributions.

g) (1pt*) Report the conditional mean and standard deviation of the **Revenues** of stores located in the city centre.

h) (2pt*+2pt**) Consider the variables **Sales** and **Revenues**.

   i. Using a suitable graph (no need to sketch it, just write the name of the plot used) briefly describe the association between the considered variables.

   ii. Using a suitable statistical summary measure (report its calculated value), comment on the strength of the linear relationships between the two considered variables.

   iii. Consider only the stores with <u>sales lower than 300</u>. What is the percentage of the stores in the sample with this characteristic?

   iv. Consider only the stores with <u>sales lower than 300</u>. Does the assessment of the strength of linear relationship between the variables **Sales** and **Revenues** for these stores change with respect to point ii.? Compare the numerical values obtained.

SKETCH SOLUTION

a) Central, as it's the category with highest absolute frequency equal to 516.

b) In the sample we find $35.59\,\%$ of the stores in the central location and $12.22\,\%$ of the stores in the hinterland location.

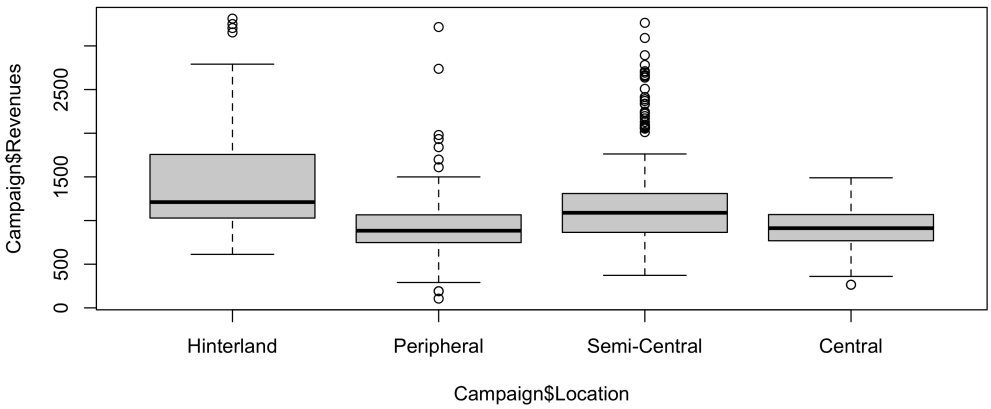
c) Among all stores in the hinterland location, $17.51\,\%$ have a loyalty score in $[80,100]$. Among all stores in the city centre location, $2.713\,\%$ have a loyalty score in $[80,100]$. Therefore the percentage of stores with the highest loyalty is much lower among stores in the city centre with respect to stores in the hinterland. This is an indication of highest fidelization of stores in the hinterland.

d) The mean of Revenues is 1057.607.

e) i. The extremes of the boxplot represent the first quartile $Q_{0.25}$ and third quartile $Q_{0.75}$ respectively. The end of the whiskers represent the minimum and maximum of the non outliers observations respectively.
ii. Using R, $Q_{0.25} = 804.430$ and $Q_{0.75} = 1202.830$. An observation is considered outlier if its value is below the lower limit $Q_{0.25} - 1.5 \cdot (Q_{0.75} - Q_{0.25}) = 207.845$ or above the upper limit $Q_{0.75} - 1.5 \cdot (Q_{0.75} - Q_{0.25}) = 1799.065$. Note that these limits **do not represent the end point of the corresponding whisker.** The lower whisker instead extends from the first

quartile to the minimum value above the lower limit, that is 264.68. The upper whiskers extends from the third quartile to maximum value below the upper limit, that is 1762.4.

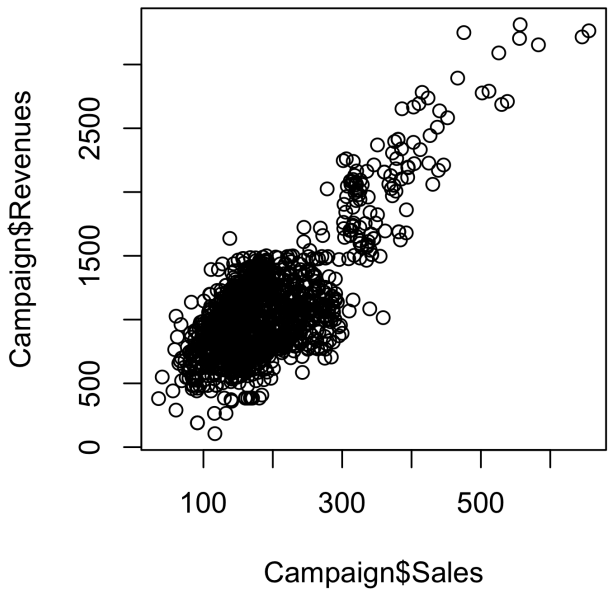
f) i. The most suitable graph for the comparison is definitely a multiple boxplot with the conditional distributions of Revenues given different values of location. Histograms for each distribution can be also used to assess skewness in particular.



ii. **Skewness.** The Hinterland box is right skewed as the upper whisker is longer than the lower whisker, the upper part of the box is wider than the lower part and it has some upper outliers and no lower outliers. The box plots for Peripheral and Semi-Central can be considered right skewed or slightly right skewed, mostly for the high number of upper outliers. This is even more evident from a histogram representation. The distribution of Revenues for the Central location is very clearly symmetrical.

g) The conditional mean and standard deviation of Revenues of stores in the city centre are 923.419 and 222.06 respectively.

h) i. To investigate the association between Revenues and Sales we produce a scatterplot.



There seems to be a positive association between the two variables i.e. the higher the sales the higher the revenues. However this association does not seem that strong for lower Sales, e.g. Sales lower than 300.

ii. To measure the strength of linear relationship we use the correlation coefficient. Its value is 0.758, which is pretty high.

iii. $91.66\,\%$ of the stores have Sales below 300.

iv. We only consider stores with Sales below 300. For these the correlation between Sales and Revenues falls to $40.43\,\%$. This is because the correlation coefficient was impacted significantly by the fewer observations with higher sales.

EXERCISE 2 (6 points)

From the dataset **Campaign** consider the variable **Loyalty**, measured in classes, that indicates the store's level of customer loyalty.

a) (1pt*) Calculate and report the relative frequencies and the frequency densities of each interval class.

b) (2pt**) We would like to target stores with relatively high loyalty value.

   i. What is the percentage of stores with loyalty higher than 75?

   ii. Explain what is the third quartile and calculate the third quartile of the variable loyalty.

c) (2pt**) Provide the mean and standard deviation of the variable loyalty.

d) (1pt*) For each value provided in question b) and c) explain whether it is exact or an estimate based on a specific assumption.

SKETCH SOLUTION

a)

	[10,20)	[20,40)	[40,50)	[50,70)	[70,80)	[80,100]
Relative Frequencies	0.04	0.22	0.24	0.30	0.12	0.08
Frequency Densities	0.004	0.011	0.024	0.015	0.012	0.004

b) i. Assuming uniform distribution within each class, the percentage of stores with loyalty higher than 75 is equal to $0.14 = 0.08 + \frac{0.12}{2}$ (relative frequency of last class plus half the relative frequency of the second to last class). Note that this correspond to the area of the histogram between the values 75 and 100.

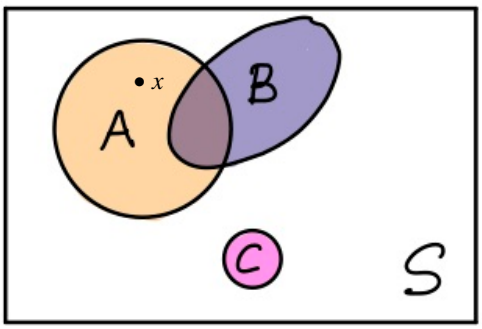
Ii. The third quartile is the value such that we have approximately $75\,\%$ of the observation below it and $25\,\%$ of the observations above it. Assuming uniform distribution within each class, it corresponds to the value of the x-axis cutting the histogram in such proportions. Its value is: $50 + \frac{0.75 - 0.04 - 0.22 - 0.24}{0.015} = 66.67$

c) Assuming uniform distribution within each class, the mean of the variable is 52.2 and its standard deviation is 19.27.

d) All of the values provided are estimates based on the assumption that data is uniformly distributed within each interval class.

EXERCISE 3 (6 points)

a) (3pt*) Let A, B and C be 3 events and x an element of $\mathcal{S}$ in A as represented in the Venn diagram below.



For each of the following statements decide whether it is TRUE or FALSE.

- $(A \cup B)^c = C$ [TRUE] [**FALSE**]
- $x \in (B \cup C)^c$ [**TRUE**] [FALSE]
- $(C^c \cap B)^c \supseteq A$ [TRUE] [**FALSE**]
- $P((A \cap C)^c) = 1$ [**TRUE**] [FALSE]
- $x \notin (A^c \cup C)^c$ [TRUE] [**FALSE**]
- $P(A^c \cap C^c) = P(B \backslash A)$ [TRUE] [**FALSE**]

b) (3pt*) For a sample $x_1, x_2, \ldots, x_n$ of numerical data, define its sample mean $\bar{x}$ and prove that the sample mean minimises the function:

$$f(a) = \frac{1}{n} \sum_{i=1}^{n} (x_i - a)^2$$

EXERCISE 4 (6 points)

In a big store, every day and independently of the other days, during the peak hour the number of customers visiting the store can be modelled as a Poisson distribution with mean parameter $\mu = 120$. (Provide the main steps in your answers to the questions below. Feel free to report your R code, although it is not strictly necessary.)

a) (1pt*) According to the model, how many customers visit the store during the peak hour in the $10\,\%$ days with the lowest number of visits?

b) (1pt*) The full capacity of the store is 150 customers and any customer in excess of full capacity cannot enter the store. What is the probability that the visits exceed full capacity during the peak hour?

c) (2pt**) If the store is open 350 days a year.

   i. What is the expected number of days in a year in which the visits will exceed full capacity during the peak hour?

   ii. What is the probability that in 3 or more days of the year the visits will exceed its full capacity during the peak hour?

d) (2pt***) (To receive points for this question, report the full R Code you use to solve it) Assume that if n clients enter the store the total amount of money they spend is normally distributed with mean $80n$€ and standard deviation $15\sqrt{n}$€.

   i. What is the expected total amount of money spent by all the clients entering the store in the peak hour?

   ii. What is the probability that the total amount of money spent by all the clients entering the store in the peak hour exceeds 12000€?

SKETCH SOLUTION

a) Define n as the number of customers arriving during the peak hour. By assumption $n \sim Poisson(\mu = 125)$. The value required is therefore the quantile of order 0.10 of n that is: $106 =$ `qpois(0.10,mu)`

b) $P(n > 150) =$ `1-ppois(150,mu)` $= 0.003552 =: p$

c) i. Out of 350, we count the number X of days in which the full capacity is exceeded, which happens with probability p, calculated before. Being the count of how many times a rare event happens, X can be modelled as a Poisson distribution with mean parameter $\mu = p \cdot 350 = 1.2432$, which corresponds to the expected number of days exceeding full capacity.

   ii. $P(X \geq 3) =$ 1-ppois(2,p*350) $= 0.1300107$.

d) There are other possible solutions and the required quantities can be calculated analytically. However the quickest one, considering the tools we have seen, is through simulation. A possible code is the following:

```
> mu = 1.2431; mean = 80; sd = 15; N = 100000
> arriving = rpois(N,mu) # simulate customers showing up for N days
> entering = pmin(arriving,150) # the store has a 150 customer capacity!
> totalspent = rpois(N,mean*entering) # total spent by
```

```
> # the customers in each of the N simulated days
> mean(totalspent) # estimated average amount spent
> mean(totalspent>12000) # estimated prob. for the total to exceed 12000
```

The final estimates should be roughly 9599 for the estimated average total amount spent and $0.27\,\%$ for the estimated probability that the total money exceeds 12000. Note that the store cannot accommodate more than 150 customers, hence every time more than 150 are simulated to arrive to the store, that amount of clients must be truncated to 150, otherwise the estimated probability in ii. is inflated significantly